Texas A&M University Invitational
January 17th, 2026

Names: _____

Team #: _____

School: _____

Heredity
Division B

Total: ___ / 48 points

Every question is worth 1 point unless stated otherwise
You have 50 minutes to complete this test
Good luck!
DO NOT WRITE ON TEST

Station 1

1. Mendel's Law of Segregation refers to which process?
    A. Separation of homologous chromosomes during meiosis I
    B. Independent assortment of non-homologous chromosomes
    C. Crossing over between homologs
    D. Equal division of cytoplasm following mitosis

2. Which scenario violates Mendel's Law of Independent Assortment?
    A. Two genes located 2 cM apart on the same chromosome
    B. Two genes on different chromosomes
    C. A monohybrid cross
    D. X-linked recessive inheritance in males

3. Which of the following produces a 9:7 phenotypic ratio?
    A. Duplicate recessive epistasis
    B. Sex-linked inheritance
    C. Dominant lethality
    D. Codominance
4. Which statement about incomplete dominance is TRUE?
    A. F_2 phenotypic ratio = 9:3:3:1
    B. Heterozygotes show a phenotype intermediate between parents
    C. Alleles blend permanently in offspring
    D. Only recessive traits appear in the F_2 generation

5. Mendel's pea plant experiments succeeded largely because…
    A. Peas are tetraploid
    B. He chose traits controlled by unlinked genes with complete dominance
    C. Peas undergo meiotic drive
    D. All traits showed intermediate expression

6. A dihybrid cross yields 9:3:3:1 phenotypes.What is the probability of getting an offspring that is homozygous recessive at one locus and heterozygous at the other?
    A. 2/16
    B. 4/16
    C. 6/16
    D. 8/16

7. A linked heterozygote has alleles in coupling: AB / ab with recombination frequency 20%. What fraction of gametes are Ab?
    A. 0.10
    B. 0.20
    C. 0.40
    D. 0.50

8. In the same situation, what fraction of gametes are AB?
    A. 0.10
    B. 0.20
    C. 0.40
    D. 0.80

Station 2

1. In rabbits, black (B) is dominant to white (b). Two black rabbits produce 96 black and 32 white offspring. What are the most likely parental genotypes?
    A. BB × BB
    B. BB × Bb
    C. Bb × Bb
    D. Bb × bb

2. Using the cross in Q1, if you randomly pick a black offspring, what is the probability it is a carrier (Bb)?
    A. 1/3
    B. 1/2
    C. 2/3
    D. 3/4

3. A plant is known to be Rr for seed shape (R = round, r = wrinkled). Its seed color genotype is unknown (Y = yellow, y = green).
    It is testcrossed with rryy and produces ~equal numbers of four phenotypes: round yellow, round green, wrinkled yellow, wrinkled green
    What is the unknown parent's full genotype?
    A. RRYy
    B. RrYy
    C. RrYY
    D. Rryy

4. For the cross RrYy × rryy, what is P(round | yellow)?
    A. 1/4
    B. 1/3
    C. 1/2
    D. 3/4

5. In RrYy × RrYy, what is the probability an offspring has exactly one recessive phenotype (i.e., round green OR wrinkled yellow)?
    A. 3/16
    B. 1/4
    C. 3/8
    D. 9/16

6. In RrYy × RrYy, the probability of wrinkled green (rryy) is 1/16. What is the probability that in 5 offspring, exactly 2 are wrinkled green?
    A. 0.032 (≈3.2%)
    B. 0.062 (≈6.2%)
    C. 0.118 (≈11.8%)
    D. 0.250 (25%)

7. An autosomal recessive disorder is caused by allele a.
   A woman is unaffected, but she has an affected sibling. Her parents are both unaffected. She has children with a man who is a known carrier.
   What is the probability their child will be affected?
    A. 1/12
    B. 1/6
    C. 1/4
    D. 1/3

8. In RrYy × RrYy, given that an offspring shows both dominant phenotypes, what is the probability the offspring is heterozygous at exactly one locus (RRYy or RrYY)?
    A. 2/9
    B. 4/9
    C. 5/9
    D. 2/3

Station 3

Dataset A (Monohybrid)

Cross Pp × Pp produced: Purple = 312, White = 108 (Total 420)

1. Expected number of purple under 3:1 is:
    A. 280
    B. 300
    C. 315
    D. 330

2. Expected number of white under 3:1 is:
    A. 105
    B. 120
    C. 140
    D. 160

3. χ^2 for Dataset A is closest to:
    A. 0.11
    B. 0.86
    C. 3.84
    D. 6.40

4. With df = 1 and χ^2 critical ≈ 3.84 (p = 0.05), best conclusion is:
    A. Reject 3:1
    B. Fail to reject 3:1
    C. Must be linkage
    D. Must be incomplete dominance

Dataset B (Dihybrid)
Observed from RrYy × RrYy (Total 498)
Round yellow 278, Round green 97, Wrinkled yellow 92, Wrinkled green 31

5. Expected wrinkled green under 9:3:3:1 is closest to:
    A. 20
    B. 31
    C. 45
    D. 62

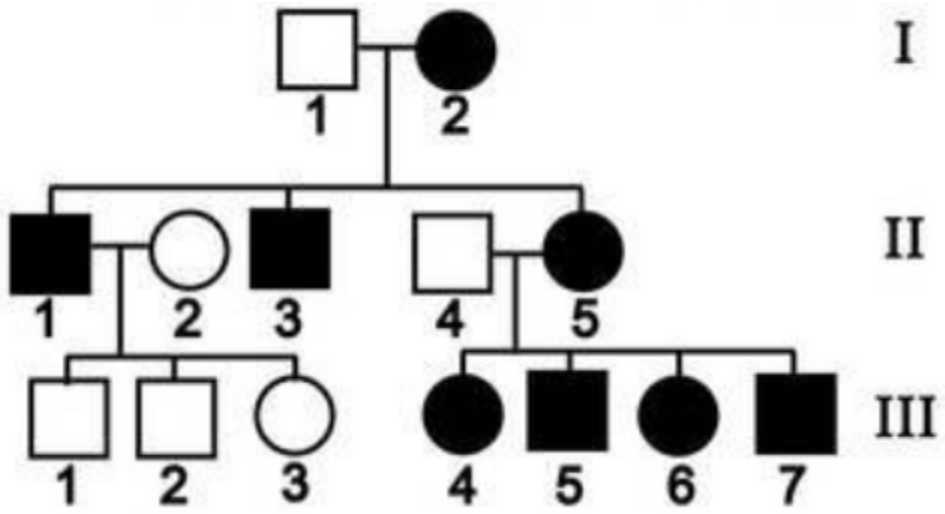
6. χ^2 for Dataset B is closest to:
    A. 0.18
    B. 2.67
    C. 7.82
    D. 15.0

Dataset C (Incomplete dominance)
Cross RW × RW observed: Red 45, Pink 98, White 57 (Total 200). Expected ratio 1:2:1.

7. Expected number of pink is:
    A. 50
    B. 80
    C. 100
    D. 150

8. Best evidence-based statement about Dataset C is:
    A. Deviations prove the trait isn't incomplete dominance
    B. χ^2 comparison can show whether deviations are consistent with sampling error
    C. A 1:2:1 ratio is impossible in genetics
    D. Only dihybrid crosses can be analyzed statistically

Station 4



1. Individual II-1 is affected, but his daughter III-3 is unaffected. This observation rules out which inheritance pattern?
    A. Autosomal dominant
    B. Autosomal recessive
    C. X-linked dominant
    D. Mitochondrial

2. Individual II-5 is an affected female, but her father (I-1) is unaffected. Assuming a single-gene trait with full penetrance, this rules out:
    A. X-linked recessive
    B. Autosomal dominant
    C. Autosomal recessive
    D. X-linked dominant

3. For this question ONLY assume II-4 is related to I-2. Individual I-2 is affected, but one of her children (II-4) is unaffected. Assuming full penetrance, this most strongly rules out:
    A. Autosomal dominant
    B. Autosomal recessive
    C. X-linked dominant
    D. Mitochondrial inheritance

4. Assume the trait is autosomal dominant and fully penetrant. Because I-1 is unaffected and II-4 is unaffected, the genotype of I-2 must be:
    A. AA
    B. Aa
    C. aa
    D. Cannot be determined

5. Assume the trait is autosomal recessive and fully penetrant. Because I-2 is affected and I-1 is unaffected but they have affected children, I-1 must be:
    A. AA
    B. Aa
    C. aa
    D. Cannot be determined

6. Assume autosomal recessive inheritance. Individual II-1 is affected. Any unaffected child of II-1 must be:
    A. AA
    B. Aa
    C. aa
    D. Either A or B

7. Assume autosomal dominant and that II-1 is heterozygous and II-2 is aa. What is the probability that all 3 of their children are unaffected?
    A. 1/2
    B. 1/4
    C. 1/8
    D. 1/16

8. Assume autosomal recessive, full penetrance, and that II-5 is affected. If II-4 is a carrier, what is the probability that all 4 of their children are affected?
    A. 1/4
    B. 1/8
    C. 1/16
    D. 1/32

Station 5

A species has 2n = 8.

1. After S phase, number of chromatids in a somatic cell is:
    A. 8
    B. 16
    C. 4
    D. 32

2. Homologous chromosomes separate during:
    A. Anaphase I
    B. Anaphase II
    C. Metaphase II
    D. Telophase II

3. Crossing over occurs primarily during:
    A. Prophase I
    B. Metaphase I
    C. Anaphase I
    D. Prophase II

4. Number of possible gamete chromosome combinations from independent assortment alone is:
    A. 4
    B. 8
    C. 16
    D. 32

5. At metaphase I, how many tetrads are aligned at the metaphase plate?
    A. 2
    B. 4
    C. 8
    D. 16

6. At metaphase II, how many chromosomes are present in each cell for this species?
    A. 2
    B. 4
    C. 8
    D. 16

7. Which statement best distinguishes mitosis from meiosis?
    A. Mitosis produces haploid cells; meiosis produces diploid cells
    B. Mitosis produces genetically identical daughter cells; meiosis produces genetically diverse haploid gametes
    C. Crossing over occurs in mitosis but not meiosis
    D. DNA replication occurs between meiosis I and meiosis II

8. Which event occurs in meiosis I but not in mitosis?
    A. Separation of sister chromatids
    B. Alignment of individual chromosomes at the metaphase plate
    C. Synapsis of homologous chromosomes to form tetrads
    D. Replication of DNA during S phase

Station 6

Case A

A double-stranded DNA molecule is 1200 bp long (so 1200 base pairs total). Its overall base composition includes 18% G.

Case B

You are told the following is the template DNA strand for part of a gene:
Template strand (read by RNA polymerase):
3′– TAC TTT GGC AGA CTT –5′

Mini codon key:
AUG = Met
AAA = Lys
CCG = Pro
UCU = Ser
GAA = Glu
UAA = Stop

Case C

A gene region is numbered on the coding strand from 1 to 900.
Forward primer binds coding strand positions 145–164
Reverse primer binds coding strand positions 612–593 (its 5′ end is at 612)

A gel is run after PCR:

- Lane 1 (ladder): 200, 300, 400, 500, 600 bp
- Lane 2 (Sample A): single band at ~470 bp
- Lane 3 (Sample B): no band
- Lane 4 (Sample C): two bands at ~470 bp and ~440 bp

1. How many hydrogen bonds are in this 1200 bp DNA molecule?
    A. 2400
    B. 2688
    C. 2832
    D. 3600

2. A phosphodiester bond forms between the:
    A. 5′ phosphate of one nucleotide and 3′ OH of the next
    B. base of one nucleotide and sugar of the next
    C. phosphate groups of adjacent nucleotides
    D. 3′ OH groups of adjacent nucleotides

3. What mRNA sequence is produced from the template strand?
    A. 5′– AUG AAA CCG UCU GAA –3′
    B. 5′– UAC UUU GGC AGA CUU –3′
    C. 5′– AUG TTT GGC AGA CTT –3′
    D. 5′– CAU UUU GGC UCU GAA –3′

4. Using the codon key, the peptide produced from the mRNA in Q3 is:
    A. Met–Lys–Pro–Ser–Glu
    B. Met–Phe–Gly–Arg–Leu
    C. Tyr–Lys–Pro–Ser–Glu
    D. Met–Lys–Pro–Ser–Stop

5. In ideal PCR, each cycle doubles the number of target DNA molecules. Starting with 1 double-stranded target molecule, how many copies are present after 6 cycles (ideal doubling)?
    A. 12
    B. 32
    C. 64
    D. 128

6. What is the expected PCR amplicon length (bp)?
    A. 448
    B. 468
    C. 488
    D. 612

7. All three PCR reactions used the same master mix (primers, polymerase, dNTPs, Mg^{2+}) and the same cycling conditions; only the DNA template added differed. Lane 3 (Sample B) shows no band. Most likely explanation is:
    A. No template DNA was added to tube B
    B. The ladder was loaded incorrectly
    C. PCR cannot amplify dsDNA
    D. Nondisjunction occurred during PCR

8. A DNA-binding repressor binds near a gene's promoter and blocks RNA polymerase binding. The most direct effect is:
    A. Increased transcription initiation
    B. Decreased transcription initiation
    C. Increased translation termination
    D. Decreased DNA replication